

Tech Stack and Decisions

Technology inventory and evidence-backed design choices.

Eurskem AI · Agentic Workflow Platform

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Derived from the live repository

Source of truth. Inventories and symbol tables are generated from the current checkout. Narrative explains observed structures and does not replace executable contracts.

Decision posture

The stack favors typed contracts, explicit infrastructure, and deployable open components. FastAPI and Pydantic keep API and node contracts near executable code. React and TypeScript make the Builder schema-driven. MongoDB stores evolving run documents, Redis coordinates ephemeral distributed behavior, MinIO protects artifacts, and Weaviate serves retrieval.

Major decisions

- Typed graph nodes over unconstrained agent loops.
- Server-resolved security scope rather than prompt-enforced isolation.
- SSE with replay for primarily server-to-client progress.
- Provider registry and fallback with cost-audited substitutions.
- HTML/CSS for PDF and native libraries for Office formats.

Dependency	Layer	Purpose
<code>fastapi</code>	Python	Typed asynchronous HTTP API.
<code>uvicorn[standard]</code>	Python	Declared backend dependency.
<code>pydantic</code>	Python	Executable schemas and configuration.
<code>pydantic-settings</code>	Python	Declared backend dependency.
<code>structlog</code>	Python	Declared backend dependency.
<code>python-jose[cryptography]</code>	Python	Declared backend dependency.
<code>cryptography</code>	Python	Declared backend dependency.
<code>argon2-cffi</code>	Python	Declared backend dependency.
<code>python-multipart</code>	Python	Declared backend dependency.
<code>motor</code>	Python	Declared backend dependency.
<code>pymongo</code>	Python	Declared backend dependency.
<code>redis</code>	Python	Coordination, events, limits, checkpoints.
<code>minio</code>	Python	Protected object storage.
<code>weaviate-client</code>	Python	Vector and hybrid retrieval.
<code>langgraph</code>	Python	Compiled graph execution.
<code>langgraph-checkpoint-redis</code>	Python	Declared backend dependency.
<code>langchain</code>	Python	Declared backend dependency.
<code>langchain-core</code>	Python	Declared backend dependency.
<code>langchain-openai</code>	Python	Declared backend dependency.
<code>langchain-anthropic</code>	Python	Declared backend dependency.

Dependency	Layer	Purpose
<code>anthropic</code>	Python	Declared backend dependency.
<code>openai</code>	Python	Declared backend dependency.
<code>pyyaml</code>	Python	Declared backend dependency.
<code>pdfplumber</code>	Python	Declared backend dependency.
<code>python-docx</code>	Python	Declared backend dependency.
<code>openpyxl</code>	Python	Declared backend dependency.
<code>python-pptx</code>	Python	Declared backend dependency.
<code>weasyprint</code>	Python	Paged HTML/CSS PDF rendering.
<code>jinja2</code>	Python	Declared backend dependency.
<code>bleach</code>	Python	Declared backend dependency.
<code>beautifulsoup4</code>	Python	Declared backend dependency.
<code>tiktoken</code>	Python	Declared backend dependency.
<code>prometheus-client</code>	Python	Declared backend dependency.
<code>mcp</code>	Python	Discoverable integration tools.
<code>paper-search-mcp</code>	Python	Declared backend dependency.
<code>mysql-connector-python</code>	Python	Declared backend dependency.
<code>httpx</code>	Python	Declared backend dependency.
<code>boto3</code>	Python	Declared backend dependency.
<code>markdown-it-py</code>	Python	Declared backend dependency.
<code>paper-qa</code>	Python	Declared backend dependency.
<code>spacy</code>	Python	Declared backend dependency.
<code>en-core-web-sm</code>	Python	Declared backend dependency.
<code>pytest</code>	Python	Declared backend dependency.
<code>pytest-asyncio</code>	Python	Declared backend dependency.
<code>@types/dagre</code>	Frontend	Declared frontend dependency.
<code>dagre</code>	Frontend	Declared frontend dependency.
<code>js-yaml</code>	Frontend	Declared frontend dependency.
<code>react</code>	Frontend	Component SPA.
<code>react-dom</code>	Frontend	Declared frontend dependency.
<code>react-router-dom</code>	Frontend	Declared frontend dependency.
<code>reactflow</code>	Frontend	Visual workflow graph.
<code>@axe-core/playwright</code>	Frontend	Declared frontend dependency.
<code>@eslint/js</code>	Frontend	Declared frontend dependency.
<code>@playwright/test</code>	Frontend	Browser tests.
<code>@testing-library/jest-dom</code>	Frontend	Declared frontend dependency.

Dependency	Layer	Purpose
@testing-library/react	Frontend	Declared frontend dependency.
@testing-library/user-event	Frontend	Declared frontend dependency.
@types/js-yaml	Frontend	Declared frontend dependency.
@types/node	Frontend	Declared frontend dependency.
@types/react	Frontend	Declared frontend dependency.
@types/react-dom	Frontend	Declared frontend dependency.
@vitejs/plugin-react	Frontend	Declared frontend dependency.
autoprefixer	Frontend	Declared frontend dependency.
eslint	Frontend	Declared frontend dependency.
eslint-plugin-react-hooks	Frontend	Declared frontend dependency.
eslint-plugin-react-refresh	Frontend	Declared frontend dependency.
globals	Frontend	Declared frontend dependency.
jsdom	Frontend	Declared frontend dependency.
postcss	Frontend	Declared frontend dependency.
tailwindcss	Frontend	Declared frontend dependency.
typescript	Frontend	Frontend contracts.
typescript-eslint	Frontend	Declared frontend dependency.
vite	Frontend	Frontend build tooling.
vitest	Frontend	Frontend tests.